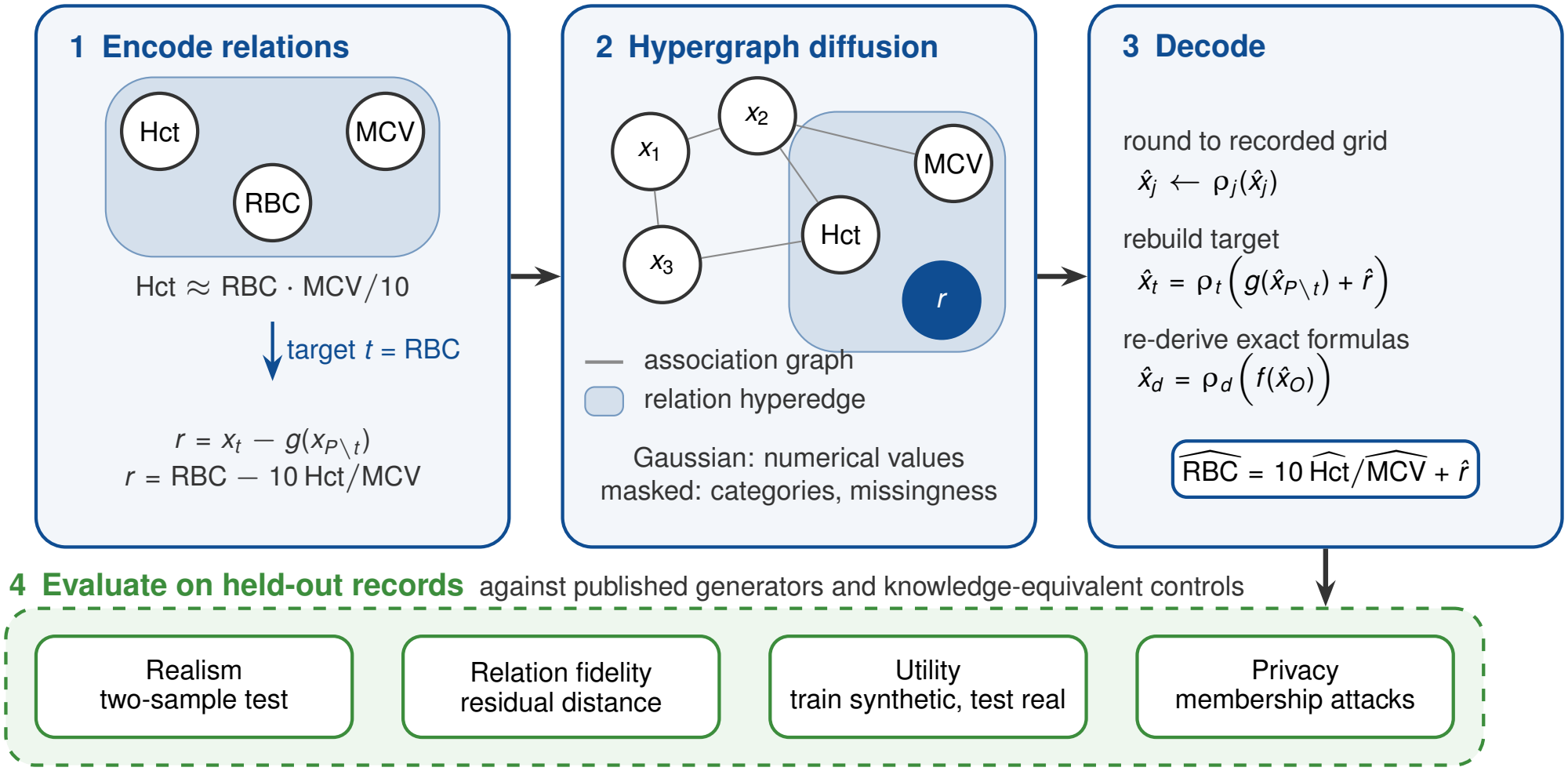

1 Encode relations
Hct
MCV
RBC
Hct ≈ RBC · MCV/10
target t = RBC
r = x_t − g(x_P\t)
r = RBC − 10 Hct/MCV
2 Hypergraph diffusion
x1
x2
x3
MCV
Hct
r
association graph
relation hyperedge
Gaussian: numerical values
masked: categories, missingness
3 Decode
round to recorded grid
x̂_j ← ρ_j(x̂_j)
rebuild target
x̂_t = ρ_t(g(x̂_P\t) + r̂)
re-derive exact formulas
x̂_d = ρ_d(f(x̂_O))
RBC = 10 Hct/MCV + r̂
4 Evaluate on held-out records against published generators and knowledge-equivalent controls
Realism
two-sample test
Relation fidelity
residual distance
Utility
train synthetic, test real
Privacy
membership attacks